

Finz Ledger Bridge

Accounting Data Pipeline, Controlled AI Classification, QuickBooks Sandbox Integration, and P&L Reconciliation

Candidate: Raina Qiu **Company dataset:** BrightFix Home Services LLC **Period:** Apr 1 - Jun 30, 2026

Source repository
github.com/RainaQiu/FinzPipeline

Public demonstration
finz-public-demo.onrender.com

Executive summary

Finz Ledger Bridge imports configurable CSV/XLSX bank exports, preserves immutable raw lineage, normalizes exact money and dates, detects duplicate records and transfer pairs, classifies transactions, routes uncertain items to human review, calculates Cash-basis P&L, creates idempotent QBO outbox intents, and supports exact account-level reconciliation.

The accounting core is deterministic. AI may propose a constrained candidate, but schema validation, the fixed account whitelist, accounting rules, and human approval remain authoritative.

Public demo disclosure: The deadline Render deployment is a zero-secret demonstration profile using an ephemeral in-memory workspace. Atlas, live Gemini, and QBO credentials are not enabled publicly. QBO Sandbox writes remain disabled, and no completed real-QBO reconciliation is claimed.

Submission contents

Item	Location / Status
Source and setup	GitHub repository and README
Working demo	Public Render URL above
Internal P&L	Monthly and consolidated statements included
QBO evidence	Read-only CompanyInfo, accounts, and empty Cash P&L verified
Reconciliation	Exact-zero workflow implemented; real posted-data result not claimed
Screen recording	Supplied separately with submission
AI usage	docs/ai-usage.md

1. Architecture and controls

Frontend and API

React provides upload, mapping, review, internal P&L, QBO, and reconciliation views. FastAPI serves both the built SPA and relative API endpoints from one HTTPS origin.

Accounting domain

Raw records are immutable. Normalized transactions use integer cents or exact Decimal money. Duplicate and transfer detection run before classification. P&L arithmetic identities are enforced.

Persistence

Repository and unit-of-work interfaces support in-memory and MongoDB implementations. The project includes authenticated local Docker MongoDB, indexes, integration tests, and an Atlas production path.

Classification

Rules run first. An optional Gemini adapter can suggest only a typed account, transaction type, confidence, and explanation from the approved 21-account chart. Low-confidence and invalid candidates require manual review.

QuickBooks boundary

OAuth state is one-time and expiring; tokens are encrypted before persistence. QBO preflight validates all 21 accounts and reuses account ID 114 for 6060 Utilities. Outbox idempotency, retry status, audit events, and stored QBO IDs protect synchronization.

Reconciliation

The application reads scoped Cash/USD QBO ProfitAndLoss responses, supports NoReportData=true, and compares each account and net profit at an exact \$0.00 tolerance.

Core data model

Model	Purpose
RawRecord	Immutable source values, hashes, and lineage
NormalizedTransaction	Canonical date, description, account, direction, currency, integer cents
ClassificationDecision	Type, counterparty, QBO account, confidence, evidence, approval, version
OutboxItem	Immutable QBO intent, idempotency key, status, retry and remote identity
ReconciliationRun	Internal, QBO, difference, status, and diagnostic candidates

2. Internally generated Cash-basis P&L;

BrightFix Home Services LLC - USD. Duplicate extras, transfers, owner activity, and fixed-asset purchases are excluded.

Period	Revenue	COGS	Gross Profit	OpEx	Net Profit
April 2026	\$98,175.00	\$31,325.00	\$66,850.00	\$44,860.00	\$21,990.00
May 2026	\$106,575.00	\$32,050.00	\$74,525.00	\$46,240.00	\$28,285.00
June 2026	\$95,525.00	\$30,475.00	\$65,050.00	\$47,145.00	\$17,905.00
Three months	\$300,275.00	\$93,850.00	\$206,425.00	\$138,245.00	\$68,180.00

Accounting checks

For every period: Gross Profit = Revenue - COGS, and Net Profit = Gross Profit - Operating Expenses. Monthly net profit sums to the consolidated \$68,180.00.

3. QuickBooks read-only verification

Field	Verified value
Company	BrightFix Home Services LLC
Environment	QBO Development Sandbox
Chart of accounts	21 active numbered accounts
Utilities	6060, existing internal ID 114
Report	ProfitAndLoss, Cash, USD, GAAP
Period	2026-04-01 through 2026-06-30
Report state	NoReportData=true

Reconciliation disclosure: QBO contains no posted challenge transactions, so its report is a valid empty report. The internal P&L; is non-zero; therefore a completed real-QBO \$0.00 reconciliation is not claimed. No mock output is labeled as real QBO evidence.

4. AI usage and independent validation

Tools used

Codex / ChatGPT: architecture translation, implementation assistance, automated tests, documentation, deployment configuration, debugging, and demo preparation.

Gemini: optional runtime candidate classifier for otherwise-unknown outflows; disabled in the zero-secret public deployment.

What AI generated

AI assisted with code scaffolding, domain and repository implementations, FastAPI and React integration, test cases, and technical documentation. Runtime Gemini may generate only a constrained candidate; it cannot alter money, approve accounting decisions, calculate final P&L, or authorize QBO writes.

Independently validated

The developer reviewed source mapping, sign conventions, P&L; inclusion rules, the fixed 21-account whitelist, and accounting arithmetic. Golden-data checks validated row counts, duplicates, transfer pairs, and consolidated totals. Read-only Intuit API Explorer queries verified CompanyInfo, all 21 accounts, 6060 Utilities ID 114, and the empty Cash-basis ProfitAndLoss response.

Assumptions

- The supplied workbook is synthetic, USD-denominated, and covers Apr 1 through Jun 30, 2026.
- The supplied 21-account chart is authoritative.
- Normalized inflows are positive and outflows are negative.
- Only the BrightFix QBO Development Sandbox is in scope; Production QBO is prohibited.
- Low-confidence and exceptional records require human approval.

Known limitations

- The public Render demo is ephemeral and loses data after restart or sleep.
- Atlas, live Gemini, and live QBO OAuth are not enabled in the public zero-secret profile.
- No real QBO transaction write or completed posted-data reconciliation is claimed.
- The shared demo has no login, tenant isolation, RBAC, or per-user data separation.
- The free service may take approximately 50 seconds to wake.

Detailed implementation, setup, safety boundaries, and delivery links are maintained in README.md and SUBMISSION.md in the repository.